

How Tough Is Location Anonymization? Re-identifying 100K Real-User Trajectories in Japan

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Context

- Trajectory data enable **unprecedented studies** across disciplines



Exploring the mobility of mobile phone users

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Unraveling Daily Human

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On the Use of h-Epidemics

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Quantifying the Mobility on Mal

Amy Wesolowski,^{1,2} Nathan Eagle,^{3,4} Abdulaziz M. Noor,^{9,10} Robert W. Snow

Social ar

Evaluating long-distance travel patterns in Israel by tracking cellular phone positions

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Collective Response of Hui Scale Emergencies

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Transportation Mode Inference from Anonymized and Aggregated Mobile Phone Call Detail Records

ibrese, Giusy Di Lorenzo, Carlo Ratti

ARTICLES

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nature physics

Modelling the scaling properties human mobility

Chaoming Song^{1,2†}, Tal Koren^{1,2†}, Pu Wang^{1,2†} and Albert-László Barabási

Individual human trajectories are characterized by fat-tailed distributions of ju relevance of continuous-time random-walk (CTRW) models for human mobility. Given the importance of human mobility, from epidemic modelling to traffic predict models that can account for the statistical characteristics of individual human human mobility, captured by mobile-phone traces, to show that the predictions of with the empirical results. We introduce two principles that govern bus self-consistent microscopic model for individual human mobility. The model also allows us to analytically predict most of the pertinent scaling exp

The Geography of Taste: Analyzing Cell-Phone Mobility and Social Events

Francesco Calabrese¹, Francisco C. Pereira^{1,2}, Giusy Di Lorenzo¹, Liang Liu¹, and Carlo Ratti¹



The temporal variation of ethnic segregation in a city: Evidence from a mobile phone use dataset

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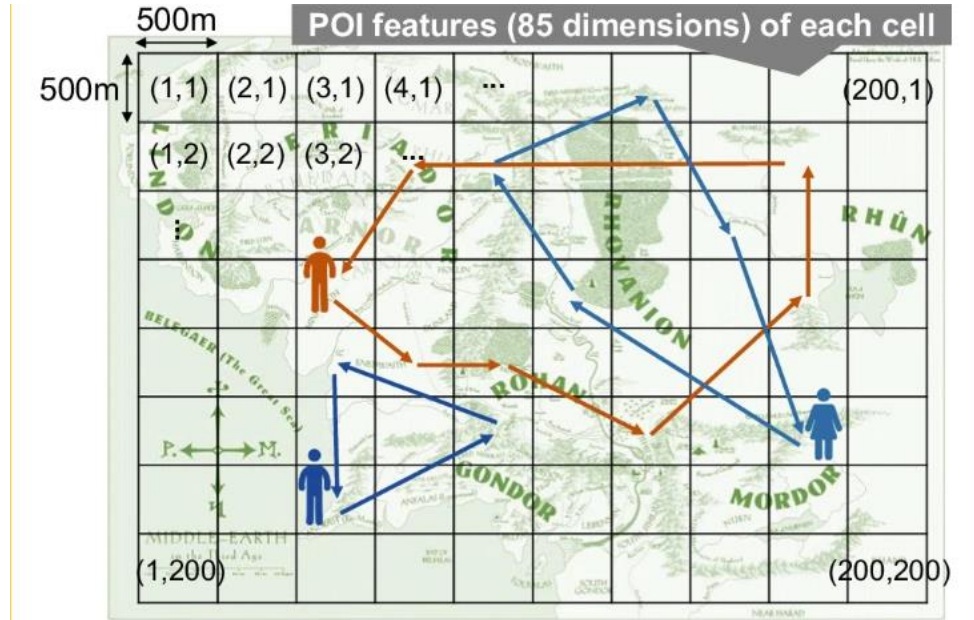
²Department of Geography, Ghent University, Krijgslaan 281, 9000 Ghent, Belgium

- Yet these personal data are also **highly sensitive**

- Can leak **private information** like residence and workplace locations, commuting habits, daily schedules, religious beliefs, or visits to sensitive places
- The collection, processing, and sharing of trajectory data are heavily regulated and very limited, which generally **curbs scientific research** and technological innovation

YJMob100K – Largest Public Release of Human Mobility

- Recent dataset^[1] of 100k user trajectories



- Has spatial generalization on a grid, time binning, and date omission
- Enough protection? Can we recover Location and Time?

Re-Identifying the Spatial Dimension

Exploiting spatial structure and density

Location Re-Identification

$$D(d) = \sum_{g \in G} P_g(d)$$


Population across
grids

Algorithm 1 Re-Identification Algorithm

```
1: Input:  $D_{\text{raw}}, D_{\text{public}}$ 
2: Output:  $C^*, T^*, S$ 
3:  $F_x(D) = \{(x, -y) \mid (x, y) \in D\}$ 
4:  $F_y(D) = \{(-x, y) \mid (x, y) \in D\}$ 
5:  $R_{90}(D) = \{(y, -x) \mid (x, y) \in D\}$ 
6:  $R_{-90}(D) = \{(-y, x) \mid (x, y) \in D\}$ 
7:  $T = \{D, F_x(D), F_y(D), F_x(F_y(D)), R_{90}(D),$   
    $R_{-90}(D), R_{90}(F_x(D)), R_{90}(F_y(D))\}$ 
8:  $S = \{\}$ 
9: for  $T_i \in T$  do
10:   for  $j = 1$  to 10 do
11:      $C_i(j) = C_{\text{corr}}(T_i(D_{\text{raw}}), D_{\text{public}}(j))$ 
12:      $S[(T_i, j)] = C_i(j)$ 
13:   end for
14: end for
15:  $(C^*, T^*) = \arg \max_{(T_i, j) \in S} C_i(j)$ 
16: Return  $(C^*, T^*, S)$ 
```

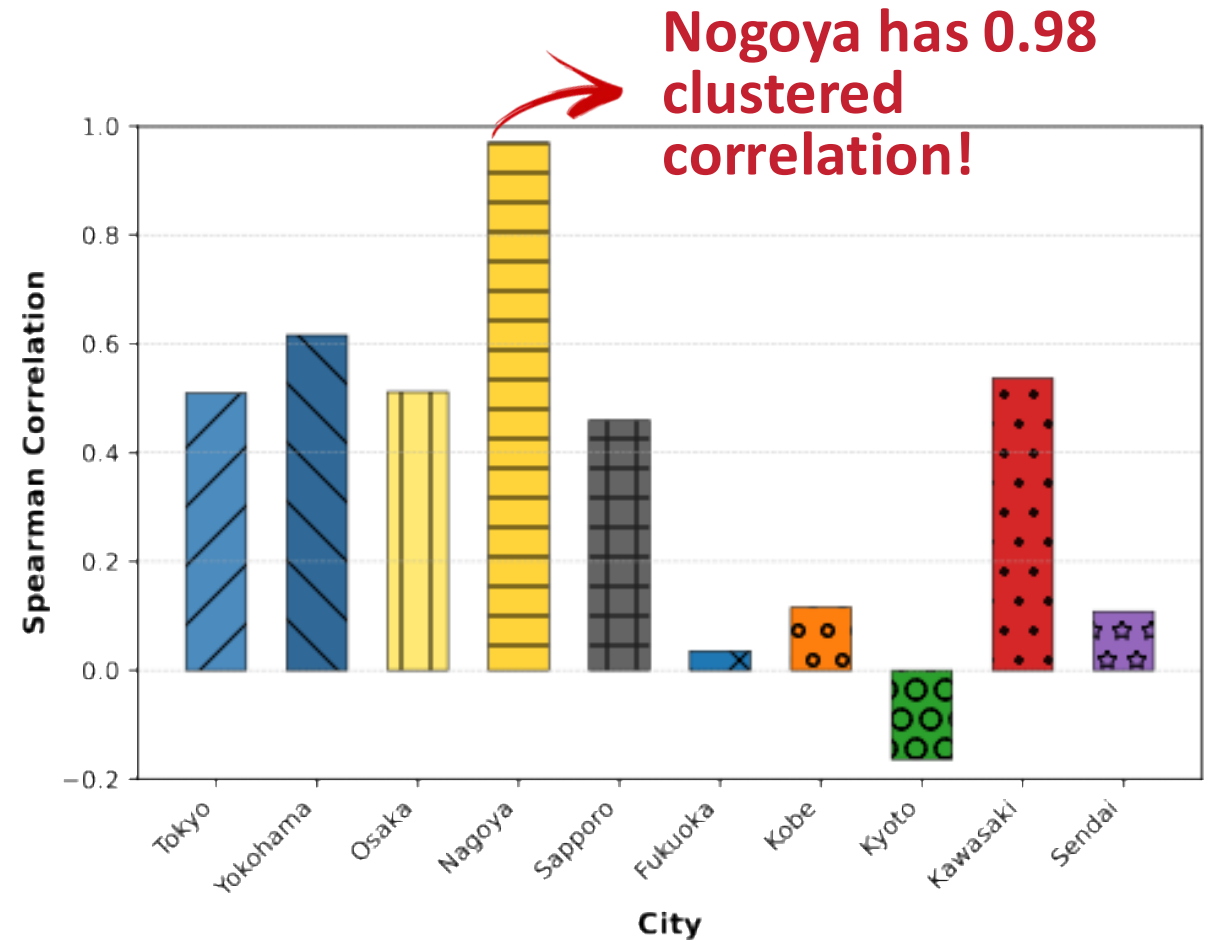
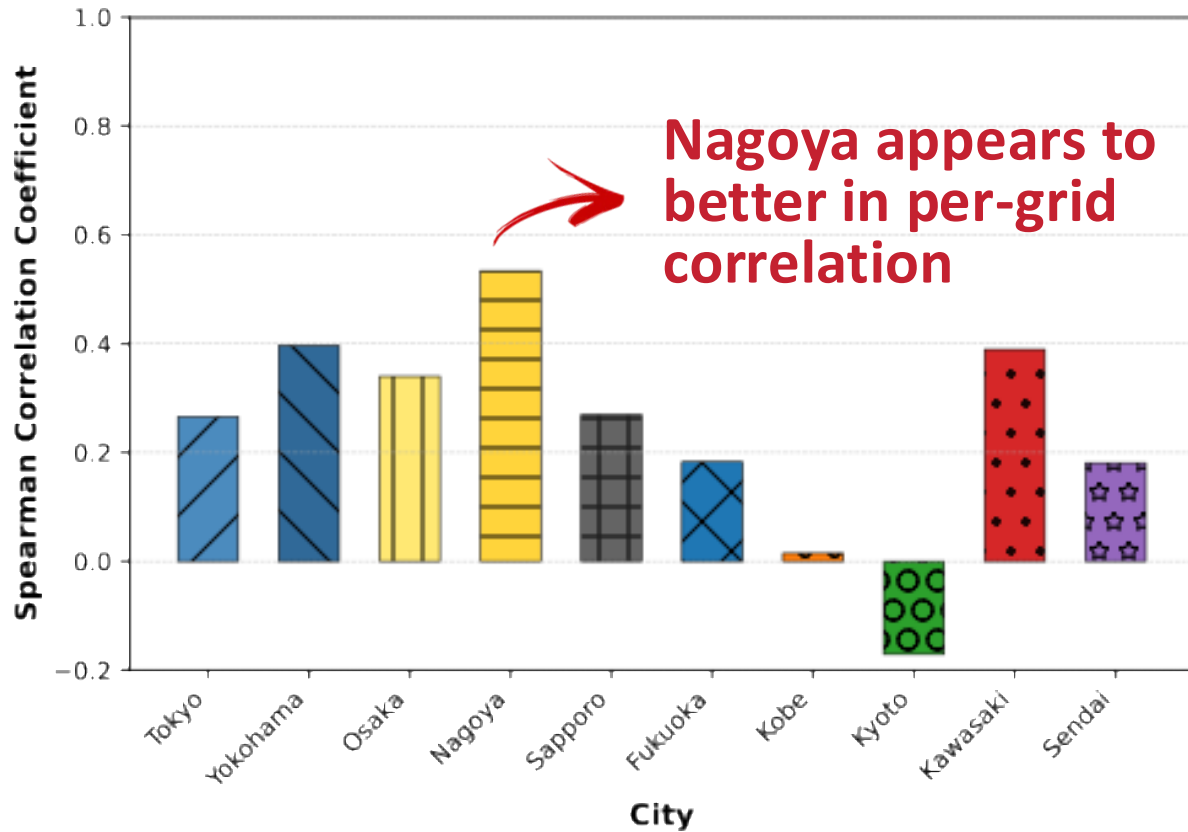
► Flip x-axis
► Flip y-axis
► Rotate +90°
► Rotate -90°

Possible transformations

Clustered correlation between
released and public population

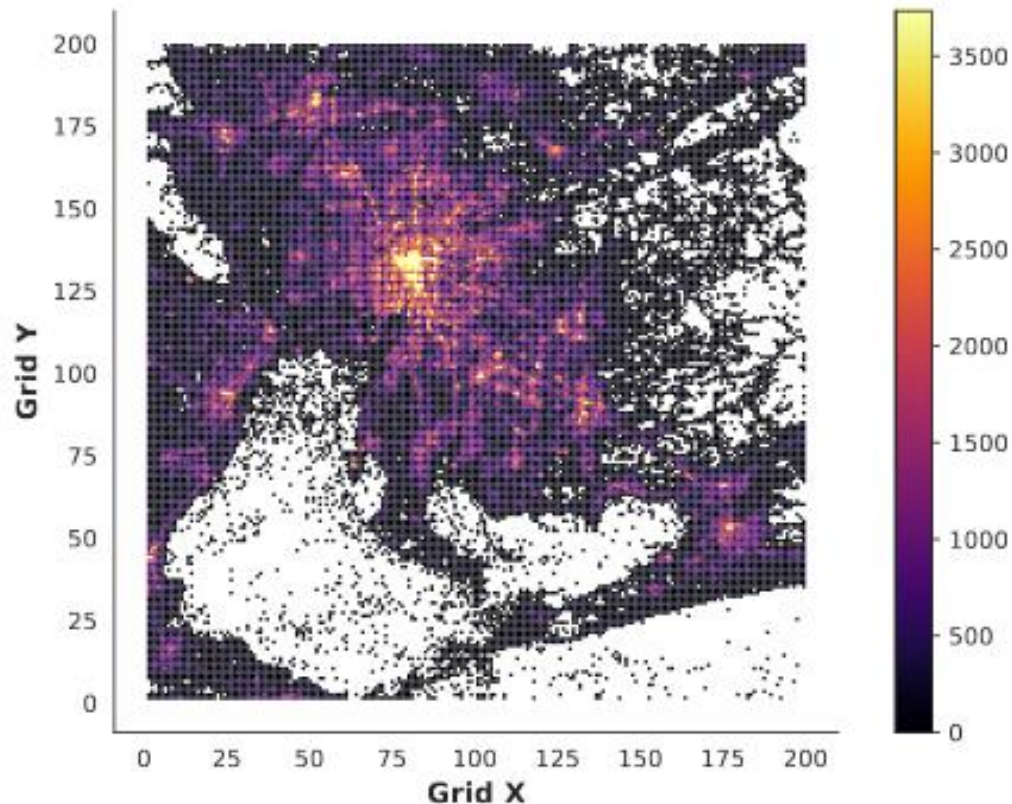


High Spatial Correlation



Confirmed Location: Nagoya

Transformed co-ordinates has visual resemblance as well!

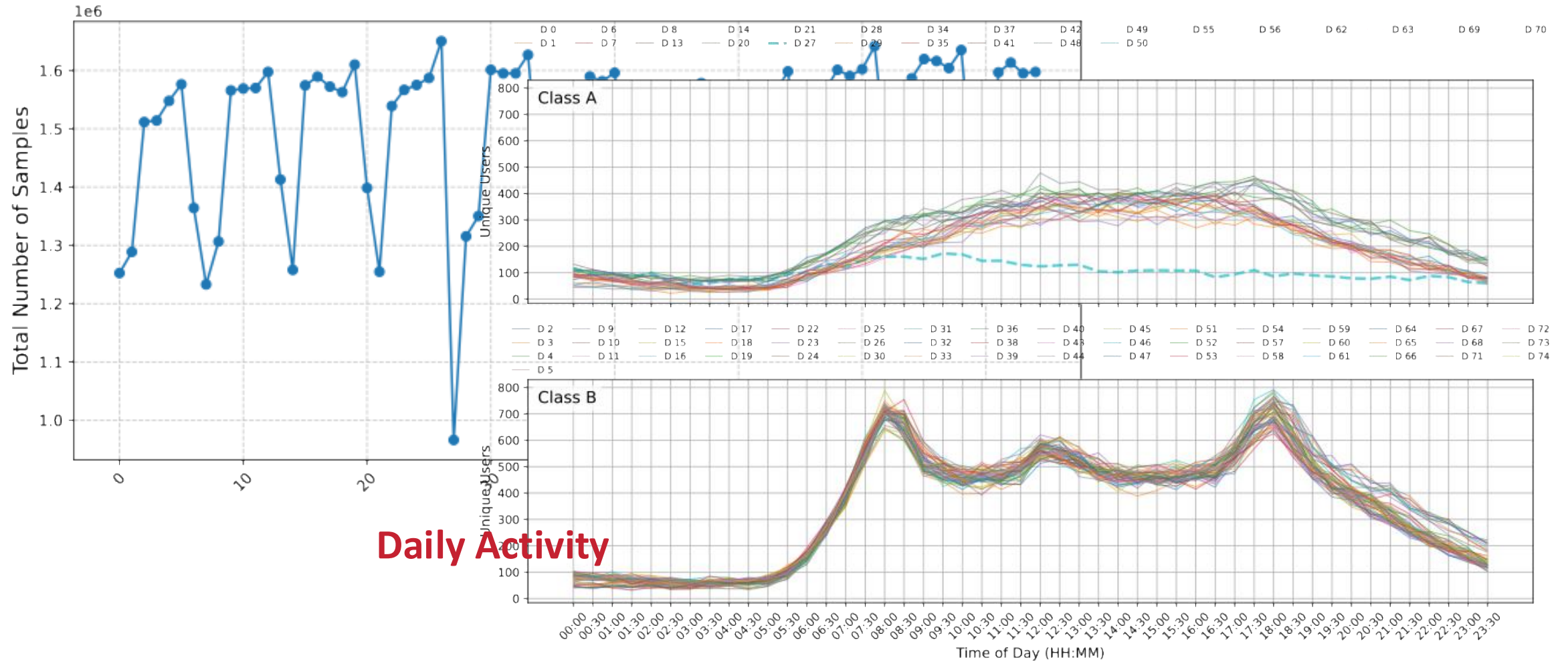


Meters level precision to the city - center by further maximizing population correlation

Re-Identifying the Temporal Dimension

Exploiting Activity Patterns, Public Events, Anomalies

Identifying Weekdays



Clear cluster of work-days vs non when looking at traffic in top residential areas

Identifying the Exact Date – Public Holidays

VS																
Day	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Class	A	A	B	B	B	B	A	A	A	B	B	B	B	A	A	
Weekday	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	
Day	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
Class	B	B	B	B	B	A	A	B	B	B	B	B	A	A	A	
Weekday	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	
Day	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	
Class	B	B	B	B	A	A	B	A	B	B	B	A	A	B	B	
Weekday	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	
Day	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	
Class	B	B	B	A	A	A	B	B	B	B	A	A	B	B	A	
Weekday	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	
Day	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	
Class	B	B	A	A	B	B	B	B	B	A	A	B	B	B	B	
Weekday	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	

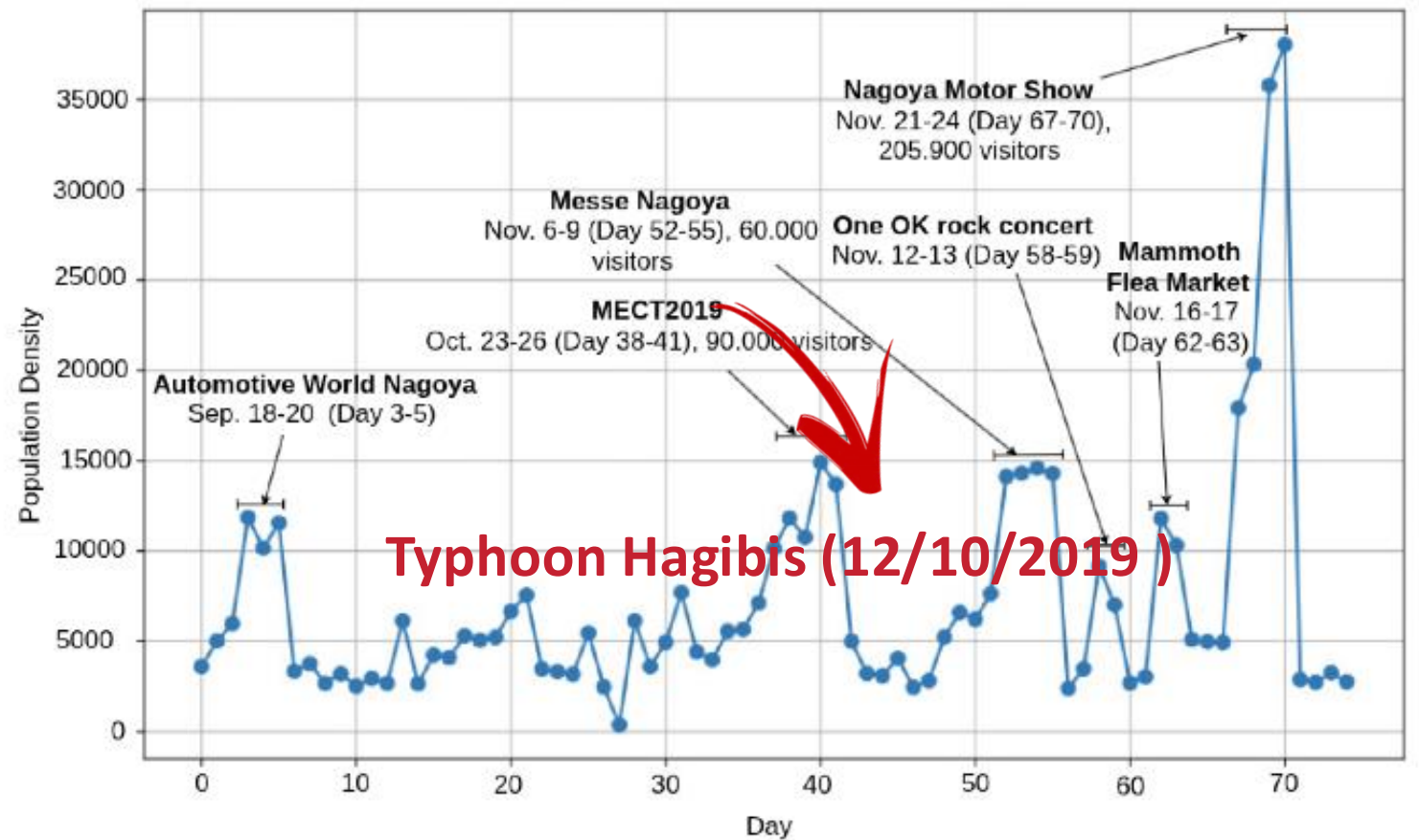
A single sequence satisfies: the dataset begins on 15 September 2019 (Day 0)!

Additional Timeline Validation

1. Activity anomaly on Day 27

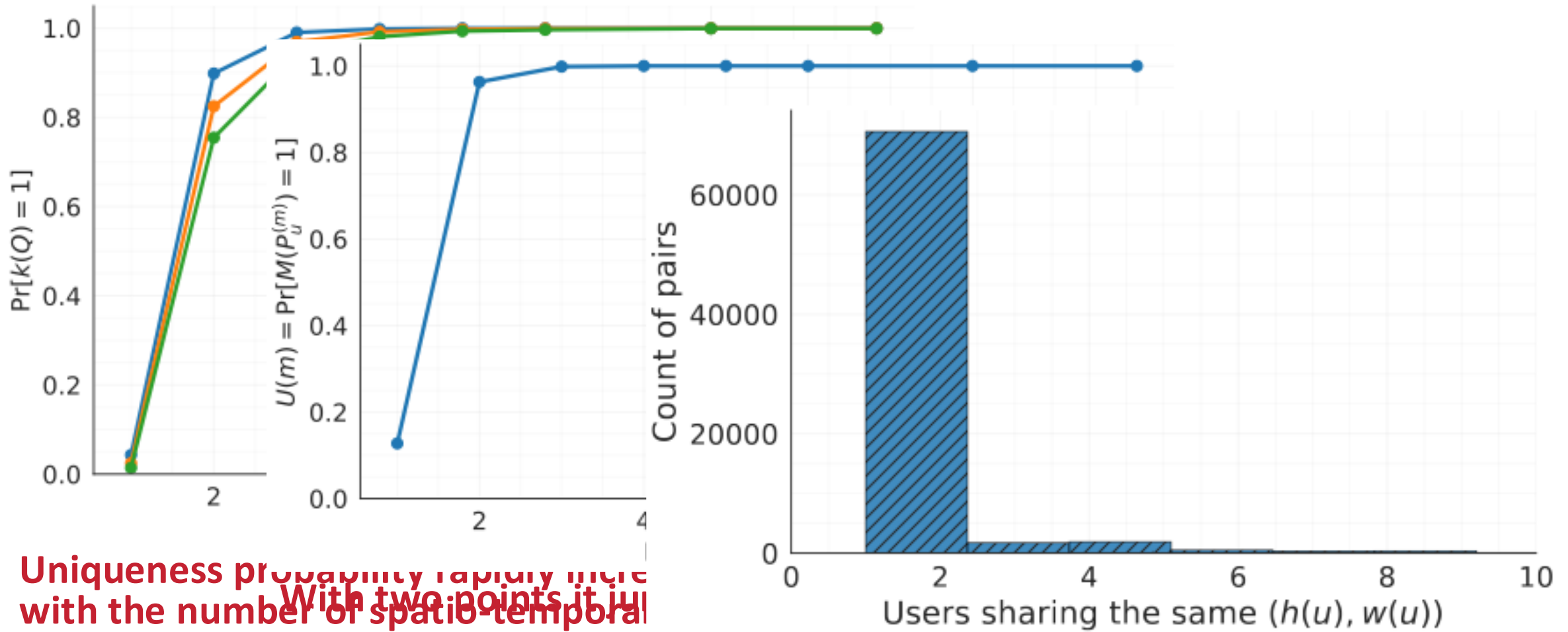


2. Event correlation at Port Messe



Privacy Risks for Users in YJMob100K

Significant Privacy Risks Found



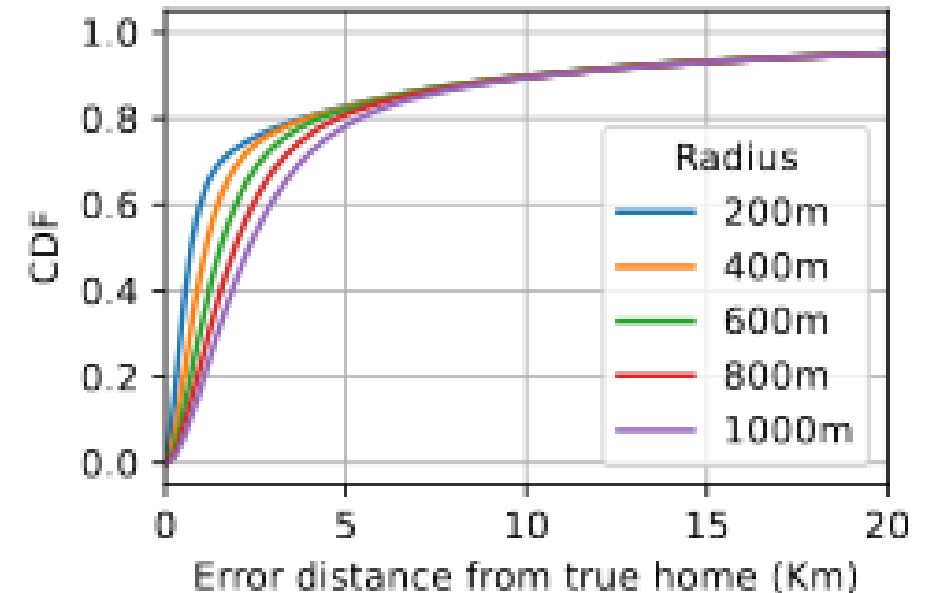
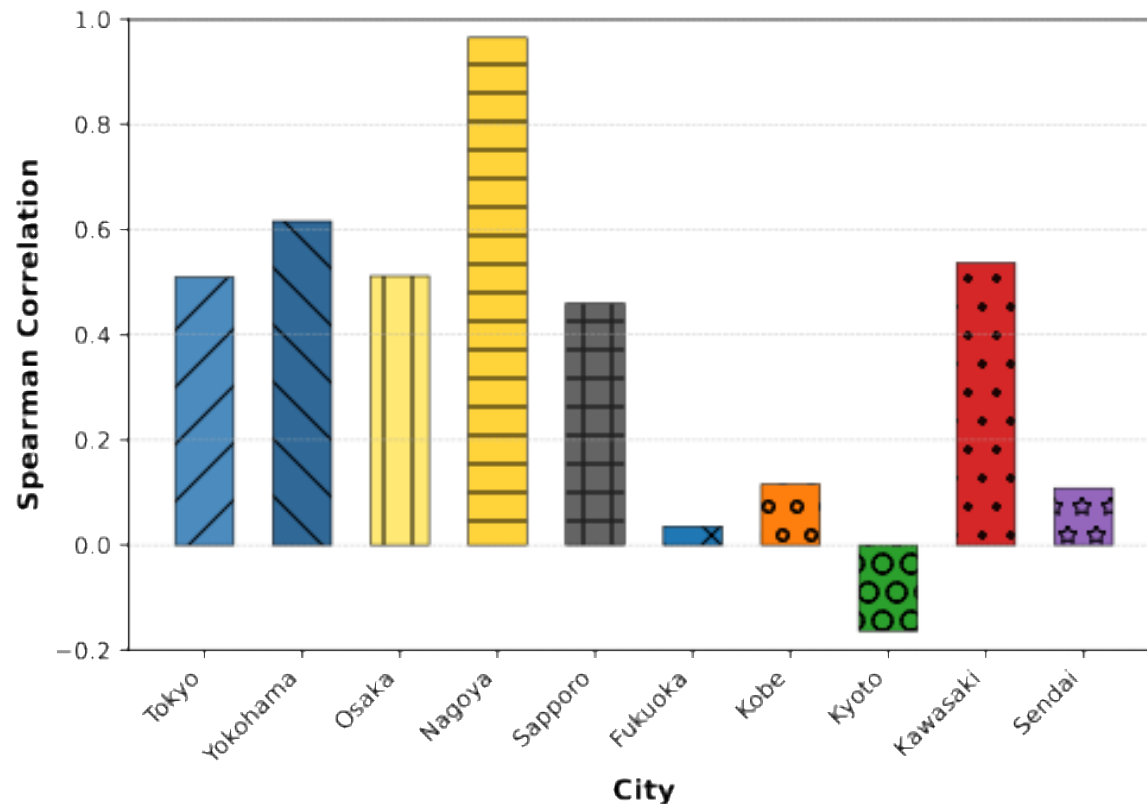
Uniqueness probability rapidly increases with the number of spatio-temporal points m with coarse temporal tolerance Δ

≈68% of users have unique home-work pair

Possible Sanitization Techniques

Geo-Indistinguishability

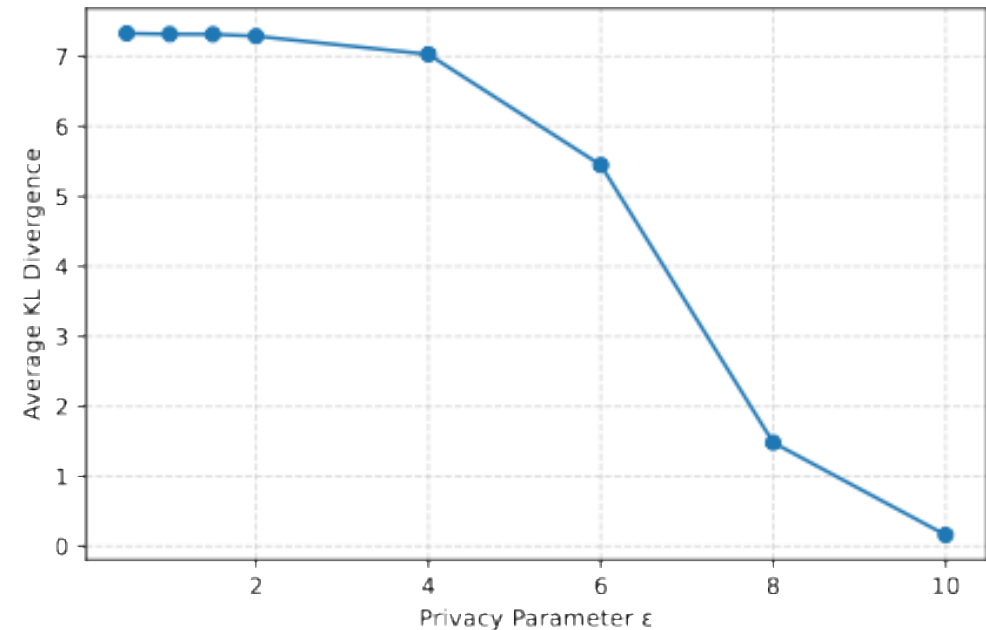
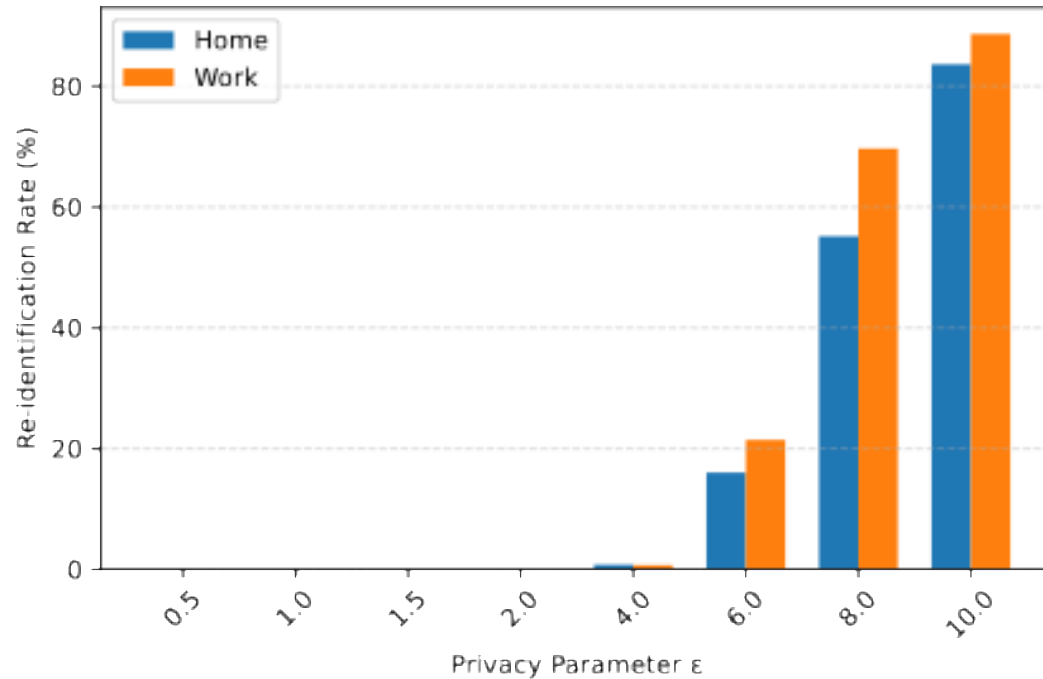
For any point x_2 within distance r of x_1 , output distributions are bounded by $l = \epsilon r$



Point-wise mechanisms does not fully obscure high-level structural information!

Generalized Randomized Response (GRR)

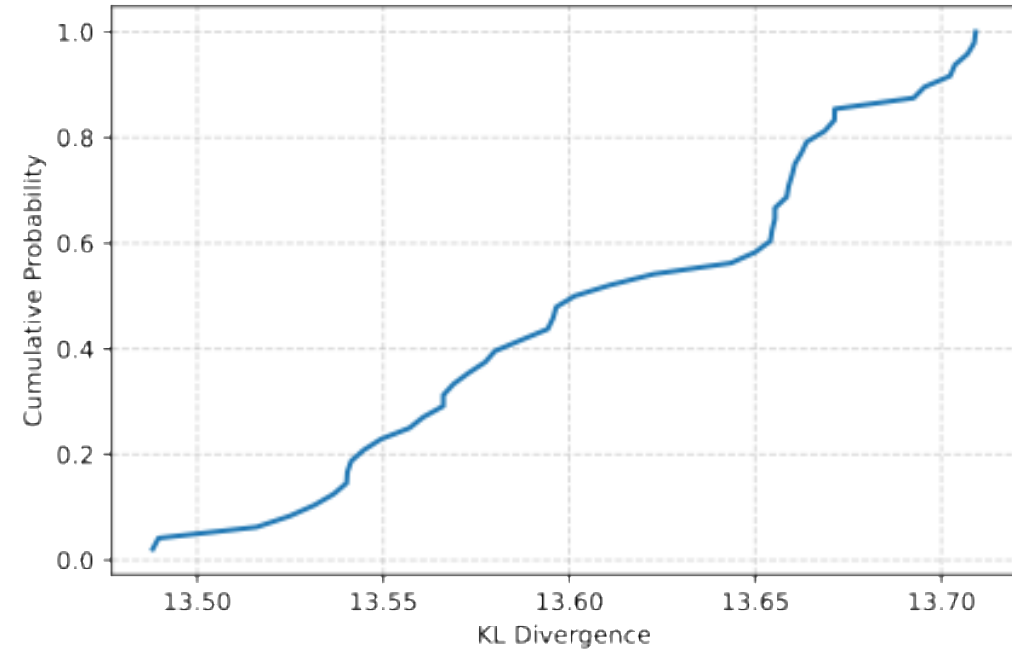
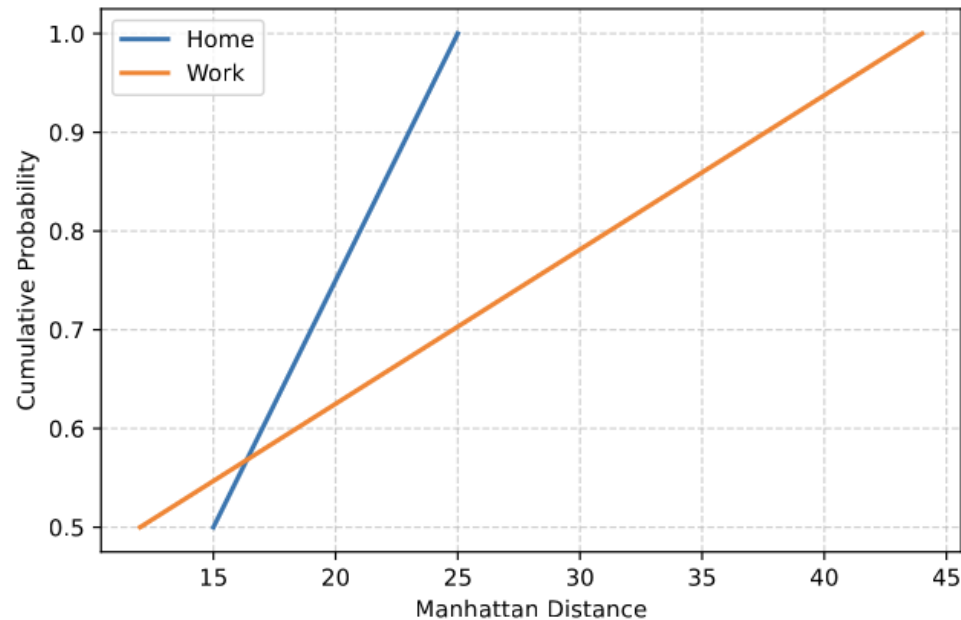
True cell with probability p , and with $1-p$ a uniformly drawn cell



Either low re-identification rate or high utility!

Spatial De-Structuring

Randomly permutes the grid-cell indices independently for each user

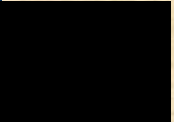


Correlation drops to 40% but Nagoya still emerges as the top candidate (by 10%) city

■ Outlook

Takeaways

- First end-to-end reversal of a large-scale anonymized mobility release
- A strong warning: as long as structural information in space and time is left intact, “anonymized” trajectories can still be traced back to individuals
- Urgent need for new privacy mechanisms to protect this structural leakage

 **Thank You! Questions?**